

MARAVEH TAPPEH, Iran, Islamic Republic of

WMO: 407210

Lat: 37.8009N

Lon: 55.9434E

Elev: 1509

StdP: 13.91

Time Zone: 3.50 (IRN)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	27.8	31.1	11.2	10.3	41.0	16.9	13.7	47.4	43.9	53.5	36.0	49.1	6.9	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	102.9	69.1	99.8	68.9	96.9	69.1	77.1	92.3	75.1	89.2	73.4	88.0	8.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	126.9	87.5	70.5	119.0	84.5	68.9	112.2	82.5	41.8	92.5	39.7	88.9	38.1	87.9	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.2	24.5	21.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	20.9	77.7	5.6	3.1	16.9	79.9	13.6	81.7	10.5	83.4	6.4	85.6	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.3	46.5	45.4	54.5	61.5	73.6	81.9	85.5	85.1	78.2	66.3	54.5	49.1
	DBStd	16.37	7.25	8.19	9.34	8.49	6.72	5.86	5.21	5.26	5.24	8.40	9.29	7.76
	HDD50	562	159	167	56	8	0	0	0	0	0	2	56	114
	HDD65	2552	573	548	348	165	8	0	0	0	1	86	328	495
	CDD50	6145	51	39	195	353	731	957	1101	1089	845	508	191	85
	CDD65	2661	0	1	22	60	274	507	636	624	396	126	13	2
	CDH74	30426	0	2	224	527	2626	6110	8303	8040	3623	906	64	1
	CDH80	15544	0	0	68	178	1168	3283	4659	4405	1455	316	12	0
(14) Wind	WSAvg	8.8	10.5	9.1	8.9	8.0	8.8	9.4	9.1	8.8	8.3	7.6	7.4	9.5
(15) Precipitation	PrecAvg	16.2	1.5	1.8	2.3	2.1	1.3	0.6	0.8	0.8	0.7	1.2	1.6	1.5
	PrecMax	24.8	3.4	3.1	4.8	4.5	4.5	2.2	3.6	2.5	3.0	3.6	3.2	2.7
	PrecMin	8.8	0.5	0.6	0.8	0.3	0.1	0.0	0.1	0.0	0.0	0.2	0.2	0.2
	PrecStd	3.7	0.7	0.7	0.9	1.1	1.0	0.6	0.8	0.6	0.7	0.7	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	71.6	88.1	91.9	100.0	105.2	107.9	105.6	98.6	95.0	81.0	73.1
		MCWB	51.8	52.8	59.7	63.7	67.6	68.3	71.5	69.3	65.9	62.4	59.1	52.0
	2%	DB	64.8	65.4	79.7	84.9	94.5	100.6	103.5	101.6	93.3	87.5	74.0	68.8
		MCWB	48.4	49.9	57.2	63.0	65.0	68.3	70.2	68.6	66.6	61.8	54.1	50.2
	5%	DB	61.3	61.3	74.4	79.9	90.8	97.2	99.8	98.3	90.4	82.6	69.5	64.4
		MCWB	46.6	47.4	55.1	60.8	64.6	68.5	70.4	68.0	66.2	60.8	52.8	48.4
	10%	DB	57.8	57.5	69.3	75.4	86.7	94.2	96.5	95.6	87.8	78.7	66.2	61.1
		MCWB	45.0	46.3	53.6	58.8	64.6	67.6	71.5	68.3	65.8	60.2	52.5	46.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	54.2	62.2	67.8	70.6	75.2	79.5	78.2	73.4	67.6	60.5	54.2
		MCDB	65.8	67.9	81.2	87.4	86.9	91.1	97.7	93.7	88.4	81.1	73.9	67.8
	2%	WB	49.6	51.2	59.1	64.7	68.9	73.2	78.1	76.1	71.6	65.7	58.2	51.9
		MCDB	61.0	61.8	75.7	80.3	85.7	91.9	94.9	90.0	84.8	77.8	70.5	63.2
	5%	WB	48.0	49.2	56.5	62.5	67.4	71.5	76.3	74.2	70.4	64.1	56.3	50.4
		MCDB	57.9	57.8	71.1	76.5	84.7	90.8	91.4	87.6	83.8	76.8	65.2	60.8
	10%	WB	46.2	47.5	54.3	60.4	65.9	70.0	74.4	72.6	69.0	62.4	54.1	48.8
		MCDB	55.9	55.0	67.4	72.6	82.1	88.7	88.8	85.9	82.5	74.7	61.9	57.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	13.0	15.7	16.3	18.4	19.5	18.3	17.3	16.0	14.9	12.9	12.0
		MCDBR	16.4	18.8	23.4	24.0	24.7	23.9	22.6	21.1	19.7	20.4	16.5	16.0
	5% WB	MCWBR	7.9	10.2	10.2	11.6	8.8	8.1	8.5	7.8	8.3	8.4	7.7	8.1
		MCWBR	15.4	16.5	20.4	21.9	21.3	21.1	18.7	16.9	16.1	16.0	13.7	14.7
(40) Clear-Sky Solar Irradiance	taub		0.356	0.400	0.467	0.524	0.528	0.524	0.546	0.491	0.464	0.456	0.392	0.357
		taud	2.223	2.089	1.923	1.838	1.867	1.892	1.852	1.975	2.013	2.003	2.151	2.226
	Ebn at Noon		254	256	249	243	245	246	239	249	247	233	237	244
		Edh at Noon	33	43	56	65	64	62	64	56	50	46	35	31
(44) All-Sky Solar Radiation	RadAvg		739	971	1310	1644	2044	2390	2299	2163	1778	1268	871	673
	RadStd		52	93	92	183	109	90	107	135	63	63	49	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)		N/A	N/A	N/A	+1.04	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page